Jonathan Hernandez Rodriguez

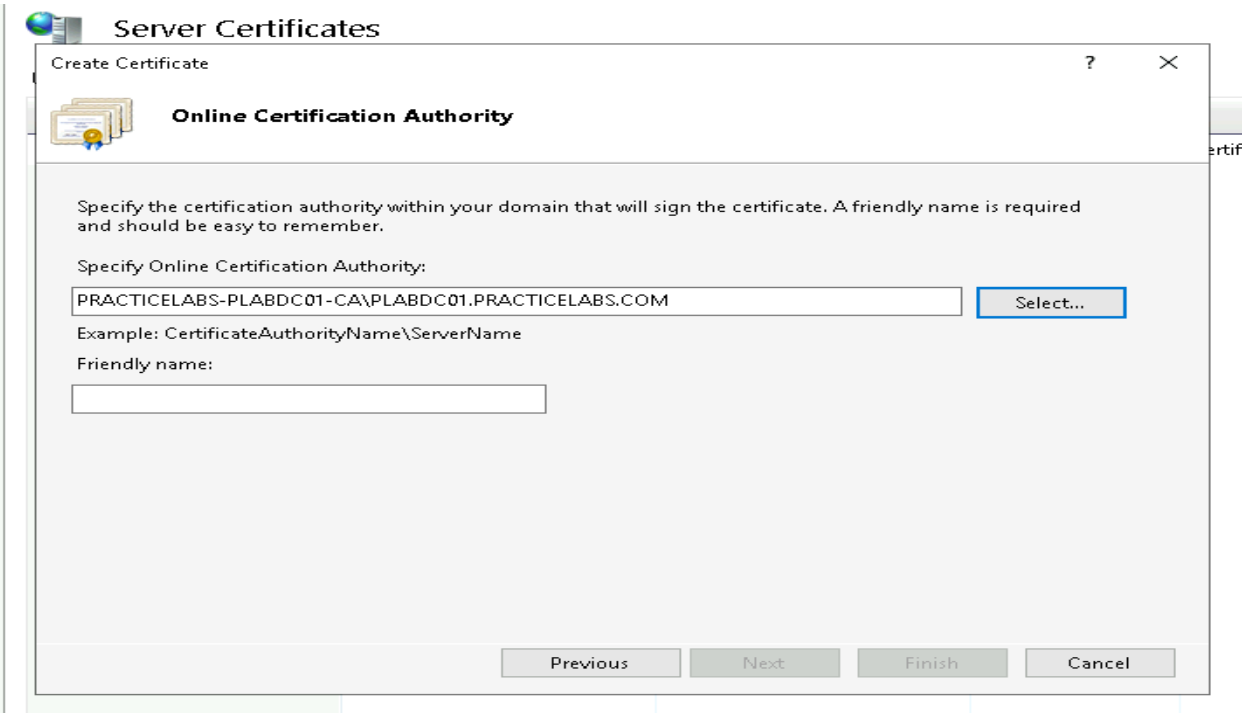
Week 4 Activity: Secure Windows 10 with Windows Defender Firewall

Task 1 - Install Web Server Role

First, I ensured the required devices were powered on. I connected to the server, right-clicked Start, and selected Windows PowerShell. In the PowerShell window, I typed the command Install-WindowsFeature Web-Server,Web-WebServer -IncludeManagementTools and pressed Enter. After waiting for the installation to complete, I closed the PowerShell window.

Task 2 - Configure a Sample Web Site and Secure Website

I opened the Server Manager Dashboard, clicked Tools, and selected Internet Information Services (IIS) Manager. In the IIS Manager window, I expanded the PRACTICELABS\Administrator node and double-clicked Server Certificates. From the Actions pane, I clicked Create Domain Certificate. In the Create Certificate window, I entered the following details: "plabdm01 web certificate" for the Common name, "Practice-Labs" for the Organization, "IT" for the Organizational unit, "London" for the City/locality, and "England" for the State/province, then changed the Country/Region to GB and clicked Next. On the next page, I selected PRACTICELABS-PLABDC01-CA and clicked OK, then finished the process by naming the certificate "plabdm01 web certificate."



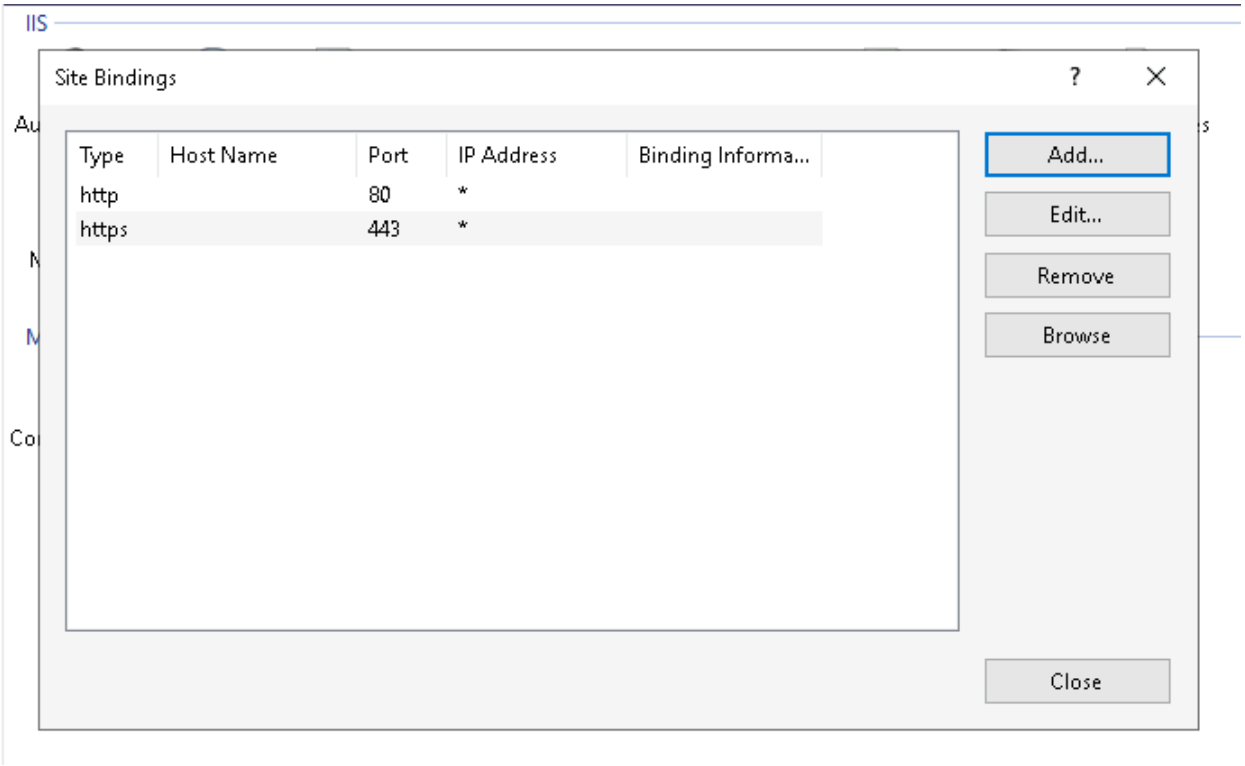
I expanded Sites, clicked Default Web Site, and selected Edit Bindings. I added a new binding for HTTPS and selected the "plabdm01 web certificate." Finally, I confirmed that both HTTP and HTTPS bindings were listed and closed the dialog box.

Task 3 - Verify Firewall Rules on the Server

I right-clicked the network icon in the system tray and selected Open Network & Internet settings. In the Settings window, I clicked the Windows Firewall link. On the Firewall & network protection screen, I clicked the Allow an app through firewall link. In the Allowed apps window, I verified that both Secure World Wide Web Services (HTTPS) and World Wide Web Services (HTTP) were selected, ensuring the necessary firewall rules were enabled for web services.

Task 4 - Test Firewall Rules on the Client

I connected to the client machine, opened Windows PowerShell, and typed Test-NetConnection -ComputerName PLABDM01 -Port 80 to test the connection to port 80 (HTTP). After confirming a successful connection, I typed Test-NetConnection -ComputerName PLABDM01 -Port 443 to test the connection to port 443 (HTTPS) and verified a successful connection for secure web traffic.



Task 5 - Change Firewall Rule on the Server

Returning to the server, I opened the Allowed apps window and cleared the checkboxes for Secure World Wide Web Services (HTTPS) and World Wide Web Services (HTTP), effectively blocking these services. From the client machine, I attempted to connect to ports 80 and 443 again using PowerShell commands, confirming the connections were now refused due to the

updated firewall rules. I also tested the connection to port 445 (SMB) and observed that it failed, confirming the rule was enforced.

Task 6 - Create an Outbound Rule on the Client

On the client machine, I opened Windows Defender Firewall, navigated to Advanced Settings, and selected Outbound Rules. I created a new outbound rule, specifying TCP port 445 (SMB) and setting the action to block the connection. I named the rule "Block 445." To verify the rule, I tested the connection to SMB on port 445 using PowerShell, observing a failure message indicating the rule was enforced. Finally, I disabled the outbound rule and tested the connection again to confirm access was restored.

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Week 4 Activity 2: Configure Connection Security Rules in Windows Defender Firewall

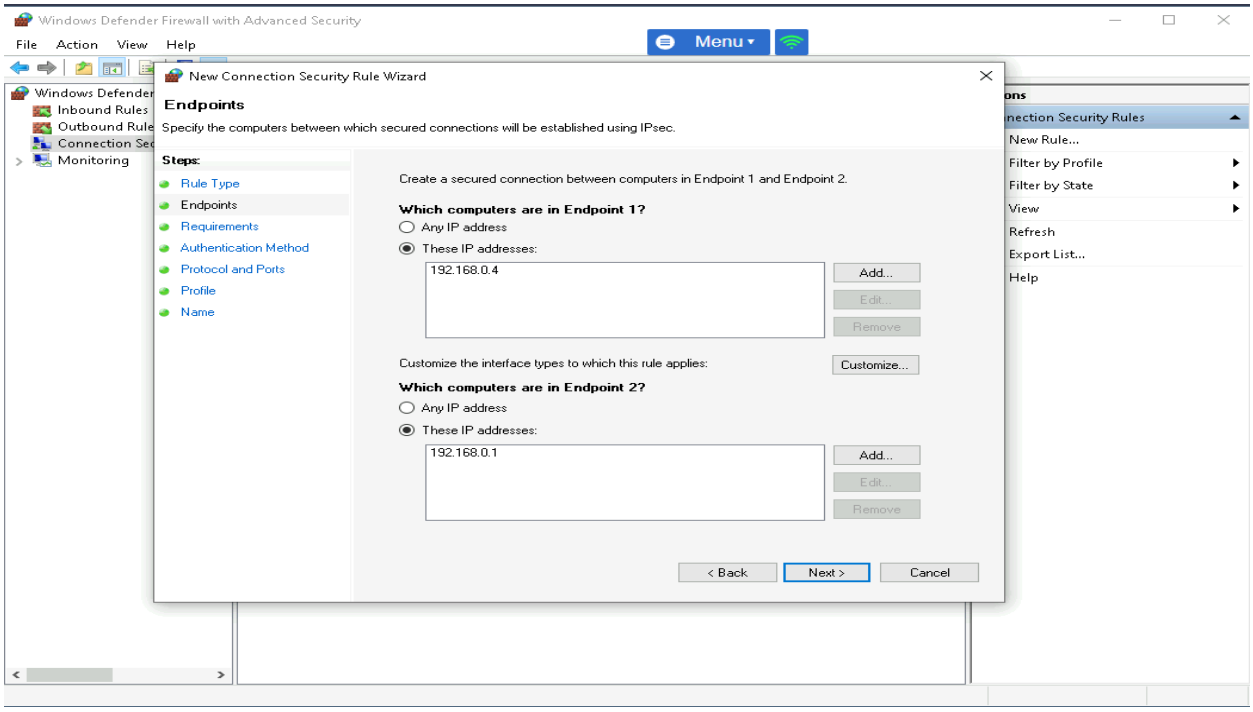
Task 1 - Create Connection Security Rules on the Client

First, I ensure the required devices are powered on. On the client machine, I click the search box and enter "advanced security." I then select Windows Defender Firewall with Advanced Security from the menu. In the Security window, I click the Connection Security Rules node, right-click, and select New Rule.

In the New Connection Security Rule Wizard, I select Custom and click Next. On the Endpoints page, under “Which computers are in Endpoint 1?”, I select These IP addresses and click Add. In the IP Address dialog box, I ensure This IP address or subnet is selected and specify the IP address 192.168.0.4. I click OK.

Similarly, under “Which computers are in Endpoint 2?”, I select These IP addresses, click Add, and specify the IP address 192.168.0.1. I click OK. I notice that Endpoint 1 and Endpoint 2 list the specified values, then I click Next. On the Requirements page, I select Require authentication for inbound and outbound connections and click Next. On the Authentication Method page, I select Computer and user (Kerberos V5) and click Next.

On the Protocol and Ports page, I select ICMPv4 and click Next. I keep the Domain, Private, and Public checkboxes selected on the Profile page and click Next. Finally, I name the rule ICMPv4 IPsec and click Finish. I keep the Windows Defender Firewall with Advanced Security window open.



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Task 2 - Enable IPsec on Client

With the Windows Defender Firewall with Advanced Security console window open on the client machine, I click the Windows Defender Firewall with Advanced Security node. In the center pane, I click the Windows Defender Firewall Properties link.

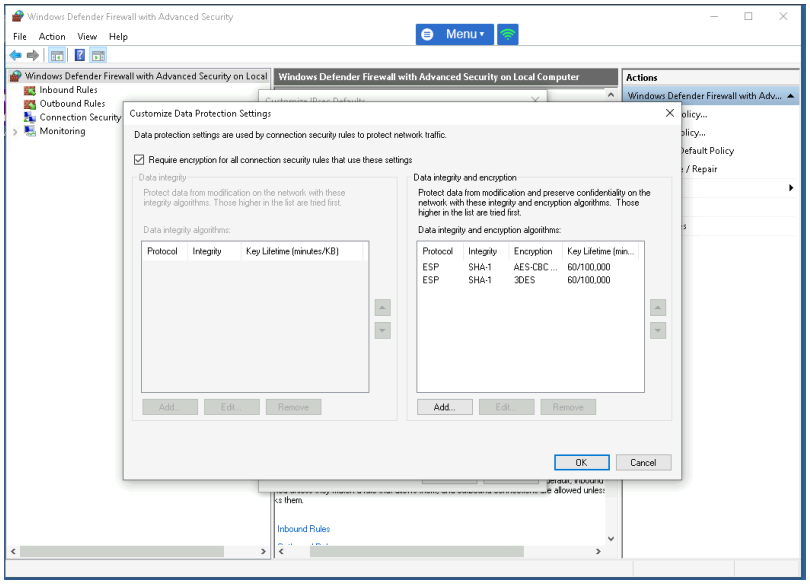
On the Windows Defender Firewall with Advanced Security on Local Computer Properties dialog box, I access the IPsec Settings tab and click Customize under the IPsec defaults section. In the Customize IPsec Defaults dialog box, I go to Data Protection (Quick Mode), select Advanced, and click Customize. I select Require encryption for all connection security rules that use these settings and click OK. I close the Customize IPsec Defaults and the Windows Defender Firewall with Advanced Security on Local Computers Properties dialog boxes.

Task 3 - Create Connection Security Rules on a Domain Controller

In the Server Manager Dashboard, I click Tools and select Windows Defender Firewall with Advanced Security. In the Windows Defender Firewall with Advanced Security window, I click Connection Security Rules, right-click, and select New Rule.

In the New Connection Security Rule Wizard, I select Custom and click Next. On the Endpoints page, I select These IP addresses under “Which computers are in Endpoint 1?”, click Add, and specify the IP address 192.168.0.1. I click OK. For Endpoint 2, I select These IP addresses, click Add, and specify the IP address 192.168.0.4. I click OK. I ensure the specified IP addresses for Endpoint 1 and Endpoint 2 are listed and click Next.

On the Requirements page, I select Require authentication for inbound and outbound connections and click Next. On the Authentication Method page, I select Computer and user (Kerberos V5) and click Next. On the Protocol and Ports page, I select ICMPv4 and click Next. I keep the Domain, Private, and Public checkboxes selected on the Profile page and click Next. I name the rule ICMPv4 IPsec and click Finish.



Task 4 - Enable IPsec on a Domain Controller

On the domain controller, I open the Windows Defender Firewall with Advanced Security console. I click the Windows Defender Firewall with Advanced Security node and, in the center pane, click the Windows Defender Firewall Properties link.

In the Windows Defender Firewall with Advanced Security on Local Computer Properties dialog box, I access the IPsec Settings tab and click Customize under the IPsec defaults section. In the Customize IPsec Defaults dialog box, I go to Data Protection (Quick Mode), select Advanced, and click Customize. I select Require encryption for all connection security rules that use these settings and click OK. I close the Customize IPsec Defaults and the Windows Defender Firewall with Advanced Security on Local Computers Properties dialog boxes. I then minimize the Windows Defender Firewall with Advanced Security console.

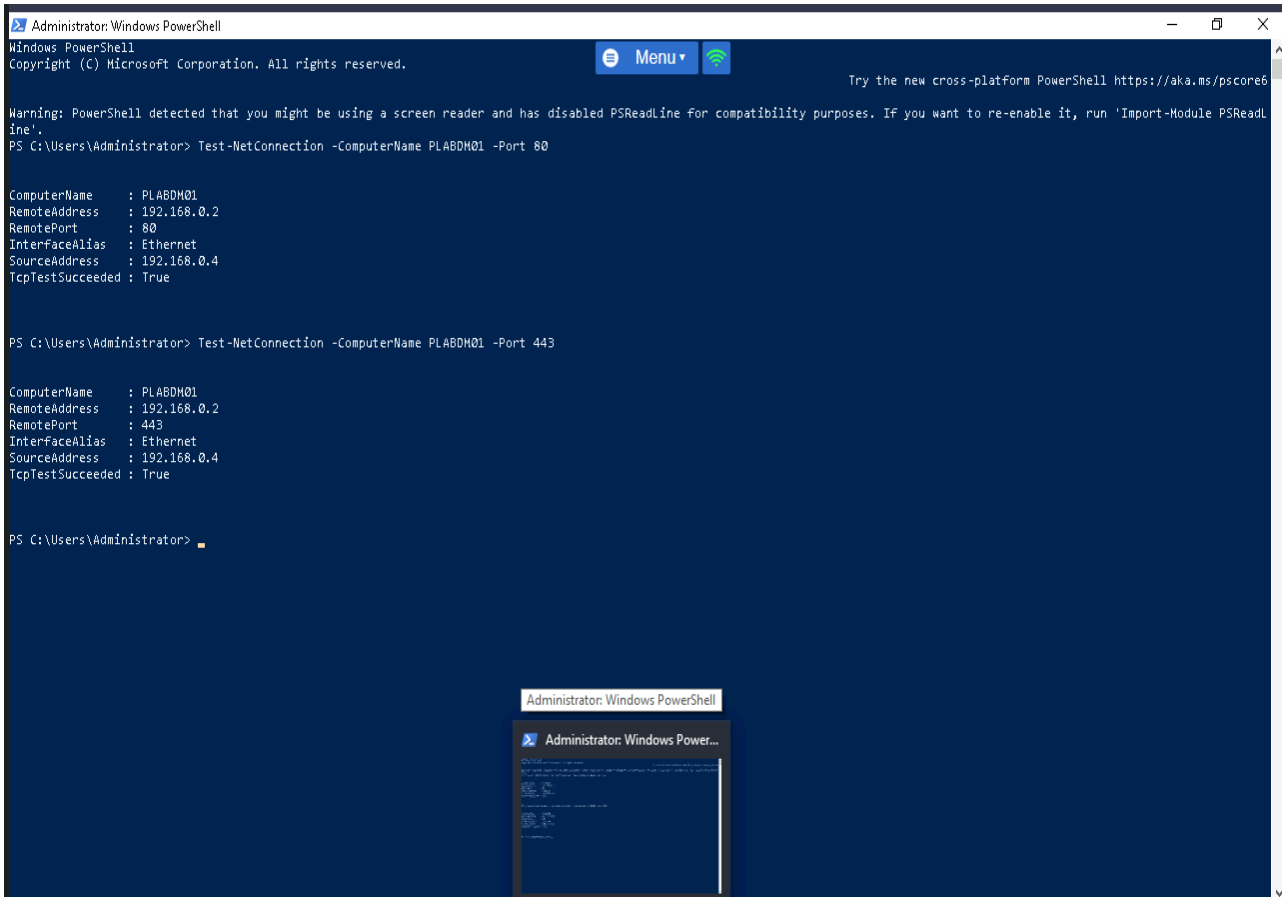
Jonathan Hernandez Rodriguez

Week 4 Activity 3: Implement Encrypting File System

Task 1 - Monitor Connection Security Rule

To monitor the connection security rule, I connect to the client machine and open Windows PowerShell by right-clicking the Start button. I enter the command ping plabdc01 -t to generate ICMP traffic between the client and the domain controller. After the domain controller responds, I minimize Windows PowerShell. Next, I restore Windows Defender Firewall with Advanced Security from the taskbar, expand Monitoring, and then expand Security Associations before clicking on Main Mode. The right pane shows information about the Main Mode, indicating successful authentication between 192.168.0.4 (PLABWIN10) and 192.168.0.1 (PLABDC01) using Kerberos V5 protocol.

I click on the IP address 192.168.0.4 in the Main Mode pane, right-click it, and select Properties. The Properties dialog box displays authentication details between the client and the domain controller, showing data signing with SHA-1. I click OK. In Quick Mode, I click on the Quick Mode pane, right-click the IP address 192.168.0.4, and select Properties again. This dialog box shows data encryption details between PLABWIN10 (192.168.0.4) and PLABDC01 (192.168.0.5). I click OK and minimize Windows Defender Firewall with Advanced Security.



Task 2 - Download and Install Network Monitor

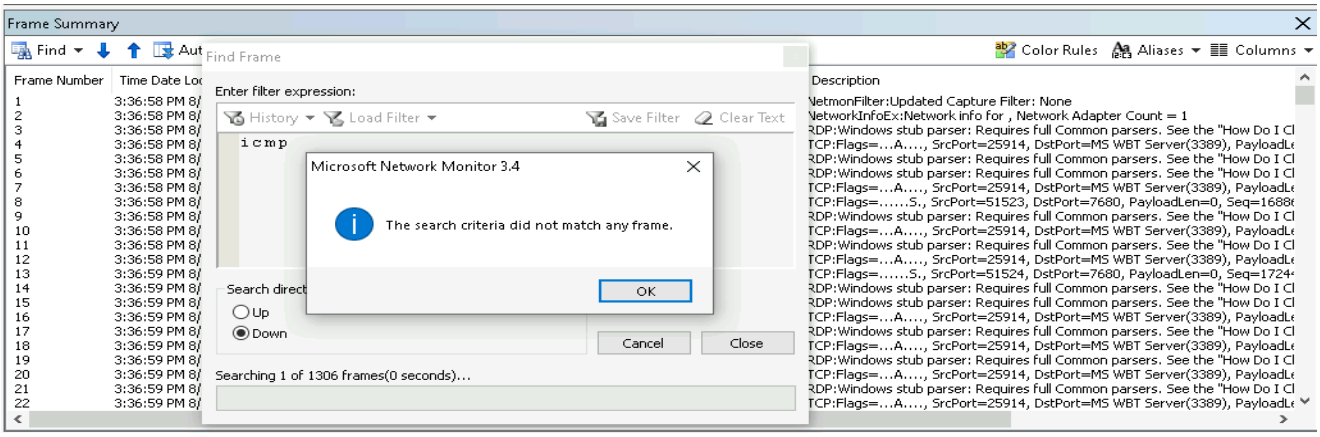
I click the Microsoft Edge icon on the taskbar of the client machine, where the intranet webpage is displayed. Navigating to the Tools and resources web page, I click Tools. On the Tools page, I scroll down to locate Windows_Server and click the Windows_Server link. On the Windows_Server path, I click NM34_x64.exe and wait for the file to download. Once downloaded, I click the Open file link that appears at the bottom-left corner of the screen. The system runs a brief security scan, and a Microsoft Network Monitor message box appears, prompting for confirmation to continue. I click Yes.

The Microsoft Network Monitor 3.4 Setup wizard starts, and I click Next on the welcome screen. I accept the terms in the License Agreement and click Next. On the Microsoft Update page, I choose not to use Microsoft Update and click Next. I select the Typical setup type and click Next, then keep the default selections on the Ready to Install page and click Install. After the installation completes, I click Finish. Network Monitor parsers install automatically during setup, and after waiting for the process to complete, I close Microsoft Edge.

Task 3 - Run Network Monitor to Observe Traffic Pattern

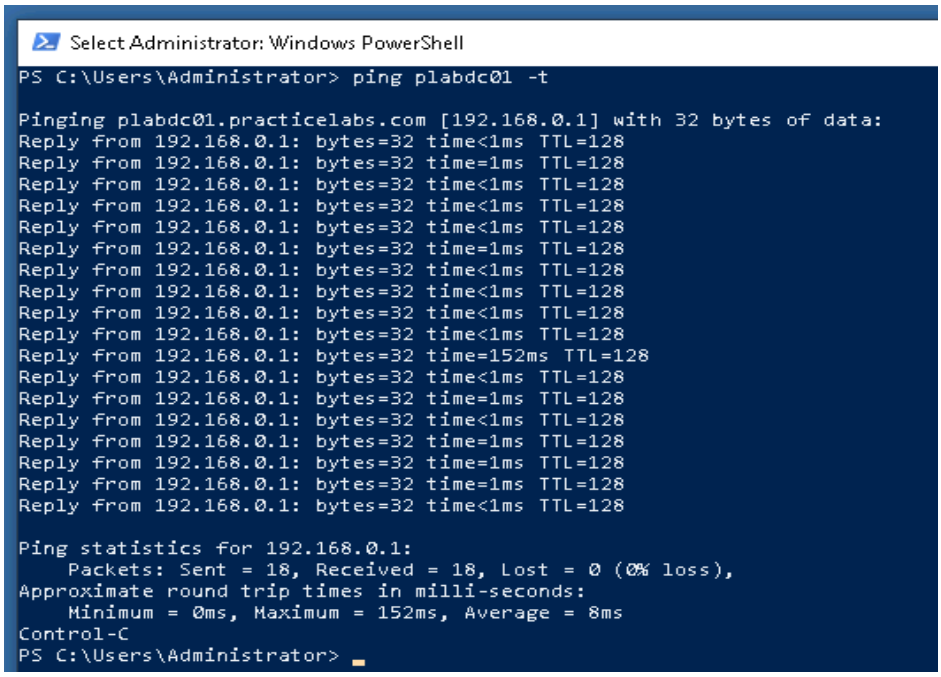
I double-click the Microsoft Network Monitor 3.4 icon on the desktop of the client machine. On the Microsoft Update Opt-in message box, I select No. In the Select Networks pane, I verify that the Ethernet checkbox is enabled. Then, in the Recent Captures pane, I click the New Capture tab. I click the Start button on the toolbar and let Network Monitor run for about one minute to capture network traffic on the selected interface. I click the Stop button on the toolbar.

To find ICMP packets, I go to the Network Conversations pane on the left and select All Traffic. In the Frame Summary pane, I click Find and select Find from the drop-down list. In the Find Frame dialog box, I type icmp and click Find. No ICMP frames appear in the network capture trace, indicating IPsec encryption. I click OK. I then type-over icmp with esp in the Find Frame dialog box and click Find. Network Monitor selects the ESP data packet found during the network capture phase. I click Close. By right-clicking one of the selected ESP frames, I point to Find Conversations and select ESP. I notice all the ESP data packets listed, showing ESP encryption of ICMPv4 packets during transmission. I minimize Network Monitor.



Task 4 - Remove Connection Security Rules

I start on the client machine by restoring Windows Defender Firewall with Advanced Security from the taskbar. I click the Connection Security Rules node on the left pane. In the center pane, I right-click ICMPv4 IPsec and select Delete. In the confirmation dialog box, I click Yes to delete the security rule and then minimize Windows Defender Firewall with Advanced Security. Next, I switch to the domain controller and repeat the steps to delete the connection security rules. Afterward, I connect back to the client machine and restore Windows PowerShell, where the ping plabdc01 -t command is still running, showing replies from the pinged server.



Task 5 - View Unencrypted Network Traffic

I restore Network Monitor from the taskbar on the client machine and click the Start button on the top toolbar. After about 30 seconds, I click Stop. In the Network Conversations pane on the left, I select the All Traffic node. In the Frame Summary section at the right, I click Find and select Find from the drop-down list. In the Find Frame dialog box, I clear the expression box and type icmp, then click Find. Network Monitor highlights ICMP data packets found during the network capture phase. I click Close and then click on the ICMP data packet to go to the Frame Details section. I notice the details of the ICMP packet, indicating its source and destination.

Jonathan Hernandez Rodriguez

Week 4 Activity 3: Implement Encrypting File System

Task 1 - Remove Windows 10 from the Domain

To demonstrate the functionality of the Encrypting File System (EFS) on my workstation, I first need to disjoin the computer from a Windows Active Directory domain. I start by ensuring that all required devices are powered on and connected to the appropriate network.

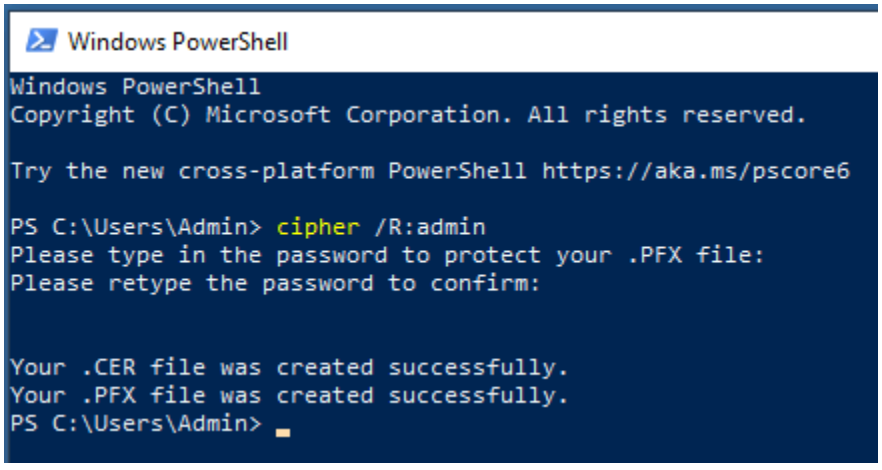
I open Windows PowerShell by right-clicking the Start menu and selecting "Windows PowerShell" from the context menu.

To remove the computer from the domain, I enter the command Remove-Computer in the PowerShell window and press Enter. When prompted to confirm, I type y and press Enter. After the command executes, I type Restart-Computer and press Enter to restart the computer and complete the disjoining process.

Task 2 - Create a Self-Signed Certificate

Since an issuing authority isn't applicable for devices in a workgroup, I need to create a self-signed certificate for the key recovery agent account. I start by reconnecting to the network. If I see a prompt about an incorrect account or password, I click OK to proceed. I sign in using the credentials .\admin with the password Passw0rd and press Enter. On the Choose privacy settings for my device screen, I click Accept and exit the Welcome to Windows screen. If I'm prompted, I click Install on any security warnings related to application installation.

I then right-click the Start menu and select Windows PowerShell. To generate a self-signed certificate, I enter the command cipher /R:admin in the PowerShell window and press Enter. I'm asked to enter a password to protect my .PFX file; I type Passw0rd and press Enter, then confirm the password by typing Passw0rd again and pressing Enter. The certificate file (.cer) and the certificate/private key file (.pfx) are created. Once this process is complete, I close Windows PowerShell.



Task 3 - Designate an EFS Recovery Agent

An EFS recovery agent is a user account that can recover encrypted files on my local computer. I need to designate a recovery agent before allowing users to encrypt their files. I start by opening the Local Security Policy console by clicking in the Type here to search box and entering secpol.msc.

In the Local Security Policy window, I expand the Public Key Policies section and select Encrypting File System. I right-click the Encrypting File System folder and choose "Add Data Recovery Agent."

In the Add Recovery Agent Wizard, I click Next on the Welcome page. On the Select Recovery Agents page, I click Browse Folders. In the Open dialog box, I expand This PC > Local Disk (C:) > Users and select the Admin folder. I choose the admin.CER file and click Open. When prompted about the self-signed certificate, I click Yes to proceed. I click Next and then Finish on the subsequent pages to complete the process. Finally, I verify that the recovery agent named Admin is listed in the Local Security Policy console before closing the window.

Task 4 - Allow Log on via Remote Desktop

To enable local users to connect to my PLABWIN10 device using Remote Desktop Protocol (RDP), I start by right-clicking Start and selecting System. On the Settings - About screen, I navigate to the left pane and click the Remote Desktop link.

On the Settings > Remote Desktop screen, I locate the User accounts section and click the link for selecting users that can remotely access this PC. In the Remote Desktop Users dialog box, I click Add. I enter everyone in the Select Users box and click OK. I confirm the addition by clicking OK again in the Remote Desktop Users dialog box and close the Settings window.

Next, I right-click Start, point to Shut down or sign out, and select Sign out. I press Enter to complete the sign-out process.

Task 5 - Create Folders, Files, and Enable EFS

To test the proof-of-concept on protecting personal folders and files with EFS, I need to create a folder and file and apply encryption to them. I start by disabling the Auto login feature. I connect to the network and select User7 on the sign-in page, entering the password Passw0rd to log in.

After signing in, I click Accept on the Choose privacy settings screen and proceed through any security warnings related to application installation. I click Maybe later on the Microsoft Edge dialog box, then open File Explorer from the taskbar.

I expand This PC, right-click the Local Disk (C:) drive, select New, and then choose Folder. I name the folder EFSTest. I open the EFSTest folder, right-click in the blank space, point to New,

and select Text Document. I name the document Protect, open it, and type: “This is a test file for EFS.” I save and close the document.

I then return to File Explorer, right-click the EFSTest folder, and select Properties. In the EFSTest Properties dialog box, I click Advanced. In the Advanced Attributes dialog box, I select the "Encrypt contents to secure data" checkbox and click OK. I confirm the attribute changes by clicking OK on the EFSTest Properties dialog box.

I notice that a padlock icon appears on the file indicating encryption. I minimize File Explorer, open Windows PowerShell, and type cd\ to exit the current folder path. To create a new folder and file, I use the commands New-Item -Path “c:\EFSTest2” -ItemType Directory and New-Item -Path “c:\EFSTest2\Protect2.txt” -ItemType File. Then, I encrypt the new folder and its contents by typing Cipher /E /S:c:\EFSTest2. I confirm the encryption status by typing cipher.

Task 6 - View EFS Permission

After applying encryption to a folder and its contents, I want to view the permissions that allow other users to access the encrypted file. I also need to check if an EFS recovery agent is available on my computer. To do this, I open File Explorer and right-click the EFSTest folder, selecting Properties.

In the EFSTest Properties dialog box, I click Advanced. On the Advanced Attributes dialog box, I click Details. Here, I see that only User7 has access to the encrypted file. The recovery certificates section lists Admin (Admin@PLABWIN10) due to the local security policy configured earlier. I click OK to close the User Access to EFSTest dialog box and close all open dialog boxes while keeping File Explorer open.

Task 7 - View Effects of Copying or Moving Encrypted Files

In this task, I demonstrate what happens to the encrypted attributes of a file after copying or moving it. I make sure I’m the owner of the file before performing these actions. In File Explorer, I click Local Disk (C:), right-click, and select New, then choose Folder. I name the folder EFSTarget.

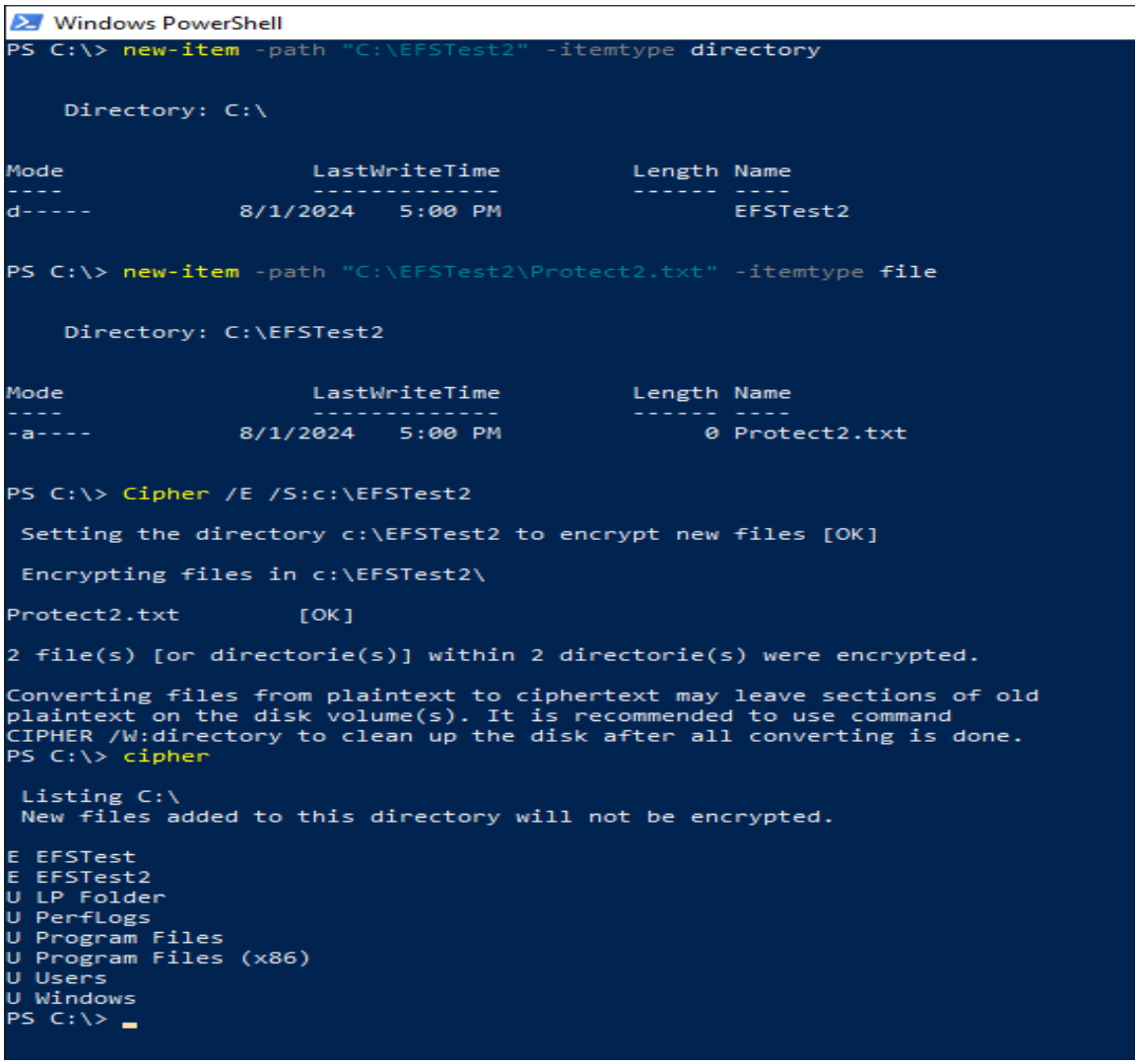
I click on the EFSTest folder, right-click the Protect file, and select Copy. I navigate to the EFSTarget folder, right-click in the blank space, and select Paste. I observe that the Protect text document retains its encrypted attribute. I then rename the copied file to Protect-Copy.

Next, I go back to the EFSTest folder, right-click the Protect text document, and select Cut. I then go to the EFSTarget folder, right-click in the blank space, and select Paste. I notice that the Protect text document retains its encrypted attribute even after being moved.

Task 8 - Decrypt the File

In this task, I use the command prompt to decrypt a folder and its files. I sign in as User7 and open Windows PowerShell. To decrypt the Protect-Copy.txt file located in EFSTarget, I type cipher /D /S:c:\EFSTarget\Protect-Copy.txt and press Enter. The text document is successfully decrypted.

For the next task, I sign out by typing shutdown /l in the command prompt and pressing Enter.



Task 9 - Recover Encrypted Files with the Recovery Agent Account

Certificates issued by a Windows Server Certification Authority can be exported in .cer and .pfx formats. The .cer format is for the certificate associated with a policy such as Encrypting File System, while the .pfx format includes the certificate with an associated private key for unlocking encrypted files.

To recover an encrypted file, I need to export the admin’s .pfx certificate. I reconnect to the network and sign in as Admin. In the Type here to search box, I type mmc and press Enter. I click Yes in the User Account Control message box to proceed.

In the Console1 window, I click File and select Add/Remove Snap-in. I choose Certificates and click Add. I verify that "My user account" is selected in the Certificates snap-in dialog box and click Finish. I click OK to close the Add or Remove Snap-ins dialog box.

In the Console1 window, I expand Certificates - Current User node, click Personal, right-click Personal, point to All Tasks, and select Import. On the Welcome to the Certificate Import Wizard page, I click Next. In the File to Import page, I type c:\Users\Admin\admin.pfx and click Next. If prompted, I enter the password Passw0rd for private key protection and click Next. I click Next on the Certificate Store page and Finish on the Completing the Certificate Import Wizard page. I confirm the successful import by clicking OK.

I then minimize the Console1 window, open File Explorer, and navigate to Local Disk (C:) > EFSTarget. I right-click the Protect file and select Properties. In the Protect Properties dialog box, I click Advanced and uncheck the "Encrypt contents to secure data" box. I click OK on both the Advanced Attributes and Protect Properties dialog boxes.

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Week 4 Activity 3: Implement Encrypting File System

Task 1 - Add Windows 10 to the Domain

In this task, I will rejoin the PLABWIN10 workstation to the PRACTICELABS.COM domain to test the relationship between a client workstation and the domain controller with a domain-wide EFS policy in place. I start by connecting to PLABWIN10 and signing in as Admin using the password Passw0rd. I then open Windows PowerShell as an administrator and enter the command Add-Computer -DomainCredential practicelabs\administrator -DomainName practicelabs.com. After pressing Enter, I provide the password when prompted and restart the computer by typing Restart-Computer in PowerShell.

Task 2 - Request EFS Recovery Agent Certificate

This task involves requesting an EFS Recovery Agent certificate from the domain controller's Certification Authority. I log in to PLABWIN10 as PRACTICELABS\Administrator with the password Passw0rd. Next, I open the Run dialog, type mmc, and press Enter to access the Microsoft Management Console. In the Console1 window, I select File and then Add/Remove Snap-in. I added the Certificates snap-in for my user account and finished the process. Then, I right-click on Personal, go to All Tasks, and select Request New Certificate. Following the prompts, I select EFS Recovery Agent and click Enroll. After completing the certificate enrollment, I click Finish to close the Certificate Installation Results page and keep Console1 open.

Task 3 - Export Certificate as .cer and .pfx Files

I will export the EFS Recovery Agent certificate in both .cer and .pfx formats in this task. With Console1 open, I expand Personal and select the Certificates folder. I right-click the Administrator certificate, choose All Tasks, and select Export. On the Certificate Export Wizard, I choose to export the private key and proceed through the wizard, setting a password for the .pfx file and specifying c:\adminefsra as the export location. After finishing the export, I perform a similar process for exporting the certificate in .cer format, specifying c:\adminefs as the location. I complete both exports and minimize Console1.

Task 4 - Enable File and Printer Sharing in Firewall

I need to ensure that the .cer and .pfx files saved on the Windows 10 workstation are accessible over the network by enabling file and printer sharing. I open Windows PowerShell as an administrator and enter the command Enable-NetFirewallRule -DisplayGroup “File and Printer Sharing” to allow incoming traffic to the file and printer sharing port. After the firewall rules are successfully enabled, I close Windows PowerShell and restart the workstation to refresh and apply the domain-wide policies.

Task 5 - Create an EFS Recovery Policy

I will use Group Policy to add a data recovery agent for EFS. I connect to the Domain Controller and open the Group Policy Management console from the Server Manager. I expand Forest: PRACTICELABS.COM > Domains > PRACTICELABS.COM and right-click Default Domain Policy to select Edit. In the Group Policy Management Editor, I expand Computer Configuration > Policies > Windows Settings > Security Settings > Public Key Policies and click Encrypting File System. I delete any existing Administrator self-issued certificates and add a new data recovery agent by selecting Add Data Recovery Agent. I browse to \\plabwin10\c$\adminefs.cer, click Open, and then Next and Finish to apply the new policy. I close both the Group Policy Management Editor and Group Policy Management console, then update the policy by running gpupdate /force in Windows PowerShell.

Task 6 - Create an Encrypted File and Verify Recovery Agent Policy

I create an encrypted file as a regular domain user and verify the recovery agent policy. I sign in to PLABWIN10 as the user jeastfield with the password Passw0rd. I created a folder named jeastfield on the Local Disk (C:) and within this folder, a new text document named Confidential. I right-click the jeastfield folder, select Properties, go to the Advanced Attributes dialog box, and select the checkbox to Encrypt contents to secure data. I click OK to apply the encryption. I then check the properties of the Confidential document and ensure that the recovery policy includes Administrator (Administrator@PRACTICELABS.COM). If the recovery agent does not appear, I restart the workstation and redo the steps. Finally, I sign out from the workstation to complete the task.

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Week 4 Activity 4: Manage Remote Desktop Access

Task 1 - Initialize a Guest Virtual Machine

To start, I'll ensure all required devices are powered on. I'll connect to the Hyper-V Manager from the Server Manager Dashboard. I'll then right-click on the guest virtual machine PLABDM03 and select Connect. When the virtual machine connection window opens, I'll click the green Start button to power up PLABDM03. Once I see the Press Ctrl+Alt+Delete screen on PLABDM03, I don't need to sign in yet. I'll manage PLABDM03 through the Server Manager Dashboard and minimize both the PLABDM03 connection window and Hyper-V Manager.

Task 2 - Install Remote Desktop Services

For installing Remote Desktop Services (RDS), I'll begin by ensuring the Server Manager Dashboard is open. I'll click Manage and select Add Servers. In the Add Servers window, I'll click Find Now, choose PLABDM03 from the list, and click the right arrow to add it. After clicking OK to close the window, I'll go to All Servers in the navigation pane. I'll right-click PLABDM03 and choose Add Roles and Features.

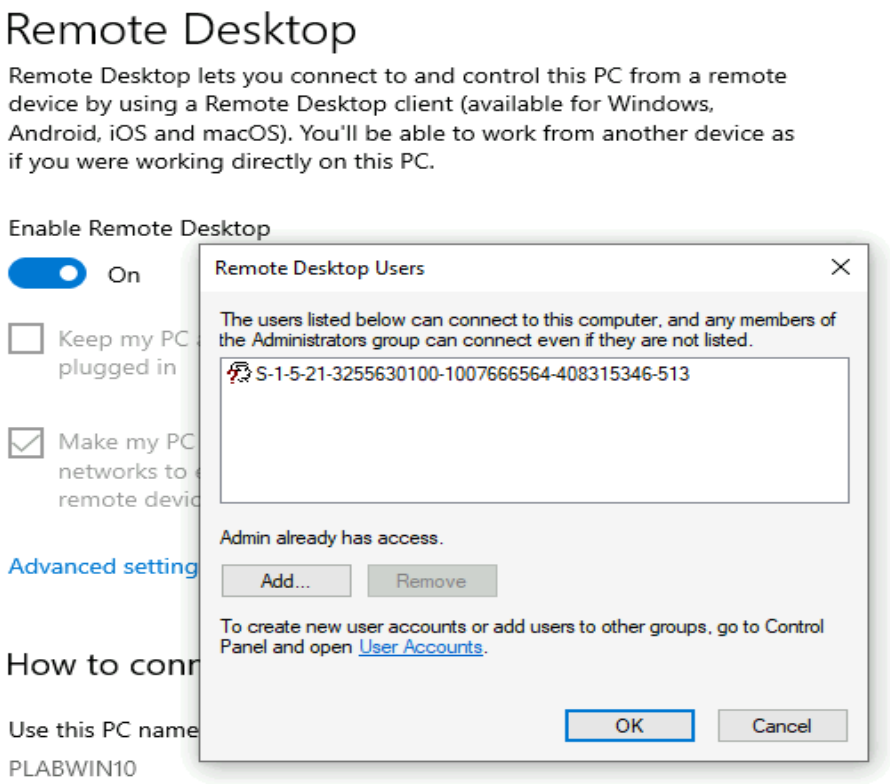
On the Add Roles and Features Wizard - Before you begin page, I'll click Next. On the Select Installation Type page, I'll choose Remote Desktop Services Installation and click Next. On the Select Deployment Type page, I'll opt for Quick Start and click Next. For the Select Deployment Scenario page, I'll select Session-based desktop deployment and click Next.

On the Select a Server page, I'll right-click under the Selected box and choose Remove. I'll then select PLABDM03.PRACTICELABS.COM and click the right arrow to add it. I'll continue with the wizard, confirm my selections, and click Install. Once the installation is complete, I'll check the Server Manager Dashboard to ensure that Remote Desktop Services is listed and verify the deployment status.

Task 3 - Install Remote Desktop Gateway

To install the Remote Desktop Gateway, I'll open the Add Roles and Features Wizard from the Server Manager Dashboard again. I'll click Next until reaching the Select Features page, where I'll check Remote Desktop Gateway and click Next. On the Remote Desktop Gateway page, I'll review the information and click Next, then continue through the wizard by confirming the installation selections and clicking Install.

After installation, I'll open the Remote Desktop Gateway Manager from the Tools menu in Server Manager. I'll configure the gateway settings according to the organization's requirements and ensure it is properly set up to handle remote connections.



Task 4 - Request Certificates for Remote Desktop Gateway

For certificate management, I'll access the Remote Desktop Gateway Manager and go to the Certificates section. I'll start the Request Certificate wizard, selecting the appropriate Web Server template. I'll complete the request process by providing necessary details to a Certificate Authority (CA) and wait for the certificate to be issued.

Once the certificate is received, I'll import it into the Remote Desktop Gateway server. I'll ensure that the certificate is correctly installed and configured by checking the Certificates section in the Remote Desktop Gateway Manager.

Task 5 - Enable the Remote Desktop Gateway Service

I'll enable the Remote Desktop Gateway service by going to the Remote Desktop Gateway Manager and checking the RD Gateway Manager settings. I'll make sure the service is set to start automatically and manually start it if needed. I'll also verify that the firewall settings allow necessary ports, including port 443 for HTTPS connections.

Task 6 - Configure Certificates for Each Remote Desktop Services Component

Finally, I'll configure certificates for each Remote Desktop Services component. In Server Manager, I'll navigate to the Remote Desktop Services section and go to the Certificates tab. I'll assign certificates to the following roles:

RD Connection Broker
RD Web Access
RD Session Host
I'll apply the changes and verify that all communication between components is secure and encrypted. To confirm everything is set up correctly, I'll test the connectivity and review any warnings or errors in the Remote Desktop Services management tools.

Jonathan Hernandez Rodriguez

Week 4 Assignment: Zero-Day Attack

The LinkedIn scraping vulnerability exploited deficiencies in LinkedIn's public profile data management and automated request handling. Specifically, the vulnerability was rooted in LinkedIn's inadequate defenses against high-volume automated scraping tools. Attackers utilized advanced scraping software to systematically extract vast amounts of information from public profiles, including sensitive details like job titles, employer names, and other professional attributes. The exploit leveraged LinkedIn's failure to effectively limit the number of requests from automated tools and to detect and mitigate unusual patterns of data access. Despite LinkedIn's existing security measures, such as CAPTCHA and rate limiting, these were insufficient to fully prevent large-scale data harvesting, exposing a critical weakness in LinkedIn's public data protection strategy.

The LinkedIn scraping vulnerability did not target specific Windows devices per se, but rather affected users across all platforms accessing LinkedIn via its web interface. This included Windows desktops and laptops, macOS systems, Linux machines, and mobile devices running iOS and Android. The exploit was related to LinkedIn's web-based infrastructure, meaning that any device capable of interacting with LinkedIn's website could potentially be affected. Since the vulnerability was in LinkedIn's handling of public data, its impact was not restricted by the operating system or device type, highlighting the widespread nature of the issue and the broad reach of the scraping tools used by attackers.

The attack was discovered through a combination of traffic analysis and monitoring of scraping activities. Cybersecurity researchers observed patterns of automated requests that were indicative of scraping operations. These activities were tracked through logs and traffic analysis, revealing a high volume of data extraction from LinkedIn profiles. Additionally, discussions and advertisements of scraping tools on dark-web forums provided further clues about the nature and scale of the attack. Researchers noticed the use of these tools to harvest public data systematically, leading to the identification of the vulnerability and a comprehensive investigation into its impact on LinkedIn's data security practices.

Microsoft was not directly responsible for mitigating the LinkedIn scraping vulnerability, as LinkedIn operates independently. The responsibility for addressing the vulnerability fell to LinkedIn itself. In response, LinkedIn took several key measures to combat the scraping activities. These included implementing enhanced bot detection technologies, refining rate-limiting mechanisms to control excessive data requests, and deploying more sophisticated algorithms to identify and block scraping attempts. LinkedIn also updated its terms of service to explicitly forbid scraping activities and pursued legal action against individuals and groups involved in these operations. Furthermore, LinkedIn engaged in ongoing efforts to improve its data protection policies and practices to better safeguard user information from unauthorized access and future scraping attempts. These actions were part of a broader strategy to enhance LinkedIn's overall security posture and protect its users from similar threats.

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